Predicting Stock Market Movements Using Political News Categories

Government changes-Based Stock Market Impact Prediction

KND_CDAP_SE_02

Project Proposal Report

S.M.K.S. SENAVIRATHNA – IT23266650

B.Sc. (Hons) Degree in Information Technology Specialized in Software Engineering

Department of Software Engineering

Sri Lanka Institute of Information Technology Sri Lanka

September 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

Name	Student ID	Signature
SENAVIRATHNA S.M.K.S.	IT23266650	

The above candidate is conducting research for their undergraduate Dissertation under my supervision.

Name	Date	Signature

ABSTRACT

Political transitions through electoral processes shape investor expectations and market behaviour in emerging economies. In Sri Lanka, Presidential elections, Parliamentary elections, Provincial Council elections, and Ministry/Cabinet transitions periodically reset the policy landscape, alter regulatory trajectories, and reshape investor confidence in the Colombo Stock Exchange (CSE). The 2024 Presidential election (21 September) saw NPP leader Anura Kumara Dissanayake secure 55.89% of votes. Immediate market responses were pronounced: the S&P SL20 reversed intraday losses to trade 2.2% higher. Within weeks, the CSE's value surged by Rs. 612 billion, with the ASPI rising 16.7% and S&P SL20 by 25%. This post-election rally followed a pre-election decline of Rs. 273 billion from the 26 July poll-date announcement, indicating election-related expectations were partially priced in ahead of the official result. The subsequent Parliamentary election (14 November 2024) delivered a landslide NPP victory with 159 seats (two-thirds majority, 62% of votes). The new 21-member Cabinet (18 November) reinforced bullish sentiment, with the CSE enjoying three consecutive months of gains. Provincial Council elections, scheduled no later than end-2025, represent another significant event category. Research covering 2000-2021 shows share prices tend to rise before elections and decline after polling dates. Ministry transitions including portfolio allocations, gazette notifications (e.g., October 2025 reshuffle reorganising the Lanka Electricity Company), and SOE reform policies further trigger sector-specific market reactions. Previous CSE research has examined election-related anomalies and political news impact on stock returns but has generally focused on measuring historical reactions rather than developing a policy-specific prediction framework integrating event-impact measurement with machine-learning forecasting. This research addresses that gap by proposing a Government Policy-Based Stock Market Impact Prediction component for the Sri Lankan context. The component will construct a structured dataset of government policy events including gazette notifications, taxation changes, import/export regulations, sector-specific decisions, and election-related announcements and associate these with historical CSE data. Event Study methodology will estimate Expected Return, Abnormal Return (AR), and Cumulative Abnormal Return (CAR) around event windows. AR, CAR, policy-event characteristics (policy category, affected sector, scope), and historical market features will inform predictive modelling using Multiple Linear Regression (interpretable baseline) and Long Short-Term Memory (time-series model). The research follows a quantitative empirical design with iterative development. Model performance will be evaluated using directional accuracy, MAE, RMSE, precision, recall, and F1-score, supported by time-series walk-forward validation to reduce data leakage. The resulting component will provide weekly market-movement predictions, policy impact indicators, and visual explanations, contributing a dedicated policy-analysis capability to the complete political-event stock prediction system integrating election, political-arrest, and government-policy components for the CSE.

Keywords: Political Event-Driven Market Prediction, Emerging Market Political Risk, Hybrid Event Study-Machine Learning Framework, Colombo Stock Exchange (CSE) Anomalies, Government Policy & Election Cycle Impact, Political Uncertainty and Investor Sentiment

ACKNOWLEDGEMENT

First and foremost, I would like to express my sincere gratitude to my supervisor, Dr. Janaka Alawatugoda, for his continuous guidance, support, patience, and valuable feedback throughout this research. His expertise and constructive suggestions have been instrumental in shaping and improving this research project.

I would also like to express my sincere appreciation to my co-supervisor, Ms. Sonali Jayarathne, for her valuable insights, constructive feedback, encouragement, and continuous support throughout the research process.

I am also grateful to my external supervisor, Dr. Emil Kumara, Senior Lecturer (Gr. I) in Finance at the Wayamba University of Sri Lanka, for his valuable expertise, guidance, and recommendations, particularly in relation to financial markets and the analysis of stock market data.

My sincere thanks also go to the academic staff of the Department of Software Engineering, Sri Lanka Institute of Information Technology (SLIIT), for providing the academic knowledge, resources, and support necessary to carry out this research.

I would also like to acknowledge the Colombo Stock Exchange (CSE) for providing access to historical market data relevant to this research.

Table of Contents

DECLARATION 2
ABSTRACT 3
ACKNOWLEDGEMENT 4
LIST OF ABBREVIATIONS 9
1. INTRODUCTION 10
1.2 Literature Review 11
1.3 Electoral Events and Stock Market Anomalies 11
1.4 Government Presidential Elections and Market Reactions 11
1.5 Parliamentary Elections and Market Responses 12
1.6 Provincial Council Elections and Market Influence 12
1.7 Elections influence on Stock Market Prediction 12
1.8 Research Gap and Influence on Proposed Research 12
1.11 Research Gap 13
1.12 Statement of Research Gap 14
2 OBJECTIVES 15
2.1 Main Objective 15
2.2 Specific Objectives 15
3 METHODOLOGY 16
3.1 Research Methodology 16
3.2 Data / Experimental Inputs 16
3.3 Baseline / Benchmark 18
3.4 Evaluation Metrics 19
3.5 Validation Strategy 20
3.6 Development Approach 20
4 HIGH-LEVEL SYSTEM ARCHITECTURE 21
4.1 Overview of Proposed Solution and Components 21
4.2 System Architecture Diagram 22
4.3 Individual Responsibility Boundary 23
5 USER REQUIREMENTS 25
5.1 Evidence from User Research 25
5.2 Functional Requirements 26
5.3 Non-Functional Requirements 27
6 BUDGET AND JUSTIFICATION 28
6.1 Hardware Costs 28
6.2 Software Licenses 28
6.3 Cloud Services 29
6.4 Human Effort Cost 30
6.5 Budget Summary 31

7 WORK BREAKDOWN STRUCTURE (WBS) 32
7.1 Task Breakdown 32
7.2 Timeline Alignment 34
7.3 Individual Responsibility 34
8 GANTT CHART 35
Summary of Evidence Meetings 48
Appendix 2 — Mandatory Proposal Evidence and Decision Log 50
Appendix 3 — AI Use Disclosure 53

LIST OF TABLES

Table 1 Tables of ABBREVIATIONS 9
Table 2 Evaluation Metrics 19
Table 3 Individual Responsibility Boundary 23
Table 4 Functional Requirements 26
Table 5 Non-Functional Requirements 27
Table 6 Hardware Costs 28
Table 7 Software Licenses 28
Table 8 Cloud Services 29
Table 9 Human Effort Cost 30
Table 10 Budget Summary 31
Table 11 Timeline Alignment 34
Table 12 Summary of meeting evidence 01 48
Table 13 Summary of meeting evidence 02 48
Table 14 Summary of meeting evidence 03 49
Table 15 Mandatory Proposal Evidence and Decision Log 52
Table 16 AI Use Disclosure 54

LIST OF FIGURES

Figure 1 High level architecture..... 22
Figure 2 WBS flow..... 34
Figure 3 Gantt chart..... 35
Figure 4 Survey result section B..... 38
Figure 5 Survey result section A..... 38
Figure 6 Survey result section C..... 39
Figure 7 Survey result section D..... 39
Figure 8 Survey result section E..... 40
Figure 9 Prototype UI Home page..... 41
Figure 10 Prototype UI News page..... 42
Figure 11 Prototype UI Political News Market Impact Predictor..... 43
Figure 12 Prototype UI Prediction Results & Scenario Analysis..... 44
Figure 13 Proof of discussing the dataset with domain expert. 45
Figure 14 Proof of discussing the dataset with domain expert. 45
Figure 15 Proof of discussing the survey with domain expert. 46
Figure 16 Proof of sharing the survey with domain expert's channel. 46
Figure 17 Proof of discussing with stakeholder. 47
Figure 18 Proof of discussing the with CSE Kandy branch..... 47

LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
AR	Abnormal Return
ASPI	All Share Price Index
CAR	Cumulative Abnormal Return
CBSL	Central Bank of Sri Lanka
CSE	Colombo Stock Exchange
DL	Deep Learning
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
MLR	Multiple Linear Regression
NLP	Natural Language Processing
RMSE	Root Mean Squared Error
SEC	Securities and Exchange Commission of Sri Lanka
SUS	System Usability Scale

Table 1 Tables of ABBREVIATIONS

1. INTRODUCTION

1.1. Background and Context

Political events and uncertainty can significantly influence financial market behaviour. Events such as elections, political arrests, government changes, and major policy decisions can affect investor confidence, trading behaviour, stock prices, market returns, and market volatility [1][2]. Therefore, understanding the relationship between political events and financial markets is important for investors, financial analysts, policymakers, and other stakeholders [3][4].

In Sri Lanka, government policy decisions are significant political events that may create uncertainty about the future economic and regulatory environment. Gazette notifications, taxation changes, import/export restrictions, monetary and fiscal policy announcements, and sector-specific regulatory decisions can directly alter the operating conditions faced by listed companies [5]. Because these announcements change investor expectations regarding costs, revenues, and future business conditions, they can lead to changes in trading activity, stock prices, market returns, and volatility on the Colombo Stock Exchange (CSE) [6][7]. Previous research on the CSE has examined the broader influence of political events on stock market behaviour and identified measurable market reactions around politically significant announcements, indicating that policy-driven events can produce observable effects on market performance [8][9].

Existing approaches such as Event Study methodology can measure the impact of policy events by analysing abnormal changes in stock returns around specific announcement dates [1][2]. However, traditional event studies mainly focus on measuring historical market reactions and do not directly provide future market movement predictions [10][11]. Similarly, general stock market prediction models may not explicitly distinguish government policy events from other political and economic factors, and gazette-based or regulatory announcements are rarely treated as a distinct, structured input to predictive models [12][13]. Therefore, this research focuses specifically on government policy events and their relationship with CSE market movements.

The proposed solution will collect and categorize government policy-related events, including gazette notifications, taxation and fiscal policy announcements, import/export regulation changes, monetary policy decisions, and sector-specific regulatory announcements. Event Study methodology will be applied to calculate Abnormal Return (AR) and Cumulative Abnormal Return (CAR) around important policy events [1][2]. These features will then be combined with historical CSE market data and used with Multiple Linear Regression (MLR) and Long Short-Term Memory (LSTM) models to predict weekly stock market movements [13][14].

Government Policy Events → Event Classification → Event Study → AR/CAR → CSE Data → MLR/LSTM → Weekly Market Movement Prediction

The research supports SDG 8 – Decent Work and Economic Growth by supporting informed financial decision-making and improving understanding of financial market behaviour. It also aligns with SDG 9 – Industry, Innovation and Infrastructure using machine learning and data-driven financial analytics. Furthermore, it aligns with the Data Science and AI Research Cluster by integrating policy-event analysis, statistical methods, machine learning, and financial market prediction [15][16].

This government policy analysis component forms one part of the broader political-event-based stock market prediction system. While other components analyse elections and political arrests, this individual component specifically investigates how government policy decisions influence the Colombo Stock Exchange and how these effects can support weekly stock market movement prediction [17][18].

1.2 Literature Review

Elections represent one of the most significant categories of political events that influence stock market behavior in Sri Lanka [19][20]. Research has systematically examined how Presidential elections, Parliamentary elections, and Provincial Council elections affect the Colombo Stock Exchange (CSE), with studies spanning from 1985 to the present providing robust empirical evidence of election-driven market anomalies [7][8][9][21].

1.3 Electoral Events and Stock Market Anomalies

Research has extensively documented the relationship between elections and stock market behavior in Sri Lanka. Karunarathna and Wijayanayake [7] conducted a comprehensive study on Colombo Stock Market anomalies due to Presidential and General Elections using daily market indices from 1988 to 2015, revealing that share prices continuously increase before elections and decrease after polling dates, with exceptions during periods of high political stability or instability. The study further identified that Banks, Finance & Insurance, Hotels & Travels, and Manufacturing sectors dominantly increased in the pre-election period, while Beverage, Food & Tobacco, Construction & Engineering, and Stores & Supplies showed no anomalies during election periods [7]. Attapattu and Gunarathne [8] examined the impact of Parliamentary and Presidential elections on stock returns and found that market sentiment formation depends on the governing party: UNP-dominant governments create negative sentiment with market recovery within days, while SLFP-dominant governments create positive sentiment, and market-friendly governments generate positive responses while less market-friendly governments generate negative responses [8]. Jayasinghe [9] examined stock returns and volatility around elections using a univariate GARCH model, producing mixed results for abnormal returns but finding strong evidence for upward movement of volatility around both Presidential and Parliamentary elections [9].

1.4 Government Presidential Elections and Market Reactions

The 2024 Sri Lankan Presidential election demonstrated significant pre-election market declines followed by substantial post-election rallies. From the 26 July poll-date announcement through August, the CSE experienced an Rs. 273 billion value wipe-outs, with the S&P SL20 dipping 6.3% and ASPI declining 4.2% [22]. Investor sentiment weakened due to uncertainty, with approximately 40% of voters undecided [22][23]. Following Anura Kumara Dissanayake's victory on 21 September, the market responded positively as the S&P SL20 reversed losses to trade 1.9% higher and the ASPI gained 1.19% [23]. Within weeks, the CSE's value surged by Rs. 612 billion, with the ASPI rising 16.7% and S&P SL20 by 25%, while market capitalization reached Rs. 5.13 trillion by December 2024 [24][25]. Historical research on four presidential elections (2005, 2010, 2015, 2019) using event study methodology revealed varied patterns: the 2005 and 2015 elections exhibited positive pre-election and negative post-election behavior, 2010 showed negative behavior in both periods, and 2019 demonstrated positive behavior throughout [10][11].

1.5 Parliamentary Elections and Market Responses

The 14 November 2024 Parliamentary election delivered a landslide victory for the NPP coalition, securing 159 seats with 61.56% of votes, followed by a 21-member Cabinet formation on 18 November [26][27]. The immediate market response was muted, with the ASPI gaining only 0.2% and S&P SL20 by 0.08% on the first trading day, as the market had already priced in expectations during the pre-election week when the ASPI gained 3.1% and S&P SL20 by 3.8% [26]. Year-to-date, the ASPI was up 25% and S&P SL20 by 29.7% [26]. Earlier research confirmed that Parliamentary and Presidential elections result in structural shifts in stock returns, with the market responding quickly to political changes [8]. An analysis revealed that overall abnormal performance existed in the days following most Presidential and Parliamentary elections, with positive reactions from the share market on most occasions [12].

1.6 Provincial Council Elections and Market Influence

Provincial Council elections, established under the 13th Amendment to the Constitution in 1987, significantly influence CSE market behavior [28]. Hulangamuwa and Dissanayake [13] examined national and Provincial Council elections from 2000 to 2021 using MLR and LSTM models, introducing a binary "election influence" variable. For ten national elections, MLR and LSTM achieved 97.01% and 87.32% accuracy with election influence, compared to 96.98% and 86.66% without it. Provincial Council elections showed similar improvements: MLR achieved 97.00% and LSTM 87.09% accuracy with election influence [13]. Recent local government elections demonstrated market volatility, with the March 2025 announcement triggering panic selling [29], and the May 2025 polls saw stocks slide 0.76% as the NPP failed to secure a majority, with the ASPI falling by 119.99 basis points [30][31][32][33].

1.7 Elections influence on Stock Market Prediction

Machine learning models can identify relationships between historical market data and external information that may influence stock prices [14][15]. The study by Hulangamuwa and Dissanayake [13] demonstrated the potential of combining election information with historical stock market data using predictor variables including open, high, and low prices and trading volume. Research on political events in emerging markets shows that uncertainty surrounding elections, government transitions, and policy changes contribute to market volatility [19][20]. Political events such as elections, leadership transitions, and policy uncertainty create constant fluctuations in the CSE [21]. Studies conclude that analyzing such events provides significant insight into market prediction and investment risk reduction [12][13].

1.8 Research Gap and Influence on Proposed Research

The literature demonstrates that Presidential elections, Parliamentary elections, and Provincial Council elections can significantly influence stock market returns, investor expectations, and market volatility in Sri Lanka [7][8][9][13]. Event Study methodology has been widely used to measure abnormal market reactions around election-related events [1][2], while machine learning has demonstrated potential for incorporating election influence into stock market prediction [13][14].

However, several gaps remain. First, while individual studies have examined specific election categories, there is limited research that systematically integrates all three election types (Presidential, Parliamentary, and Provincial Council) within a unified analytical framework for the CSE [13][21]. Second, existing election studies mainly focus on measuring historical

market reactions and do not sufficiently integrate AR/CAR analysis with machine learning-based weekly market prediction [1][2][13]. Third, different stages of the electoral process such as election announcements, campaign periods, polling dates, and result announcements are not consistently represented as structured event categories [7][8][9]. Fourth, existing financial platforms do not provide a dedicated CSE-focused election impact analysis and prediction component [26][27].

Based on these gaps, the proposed research will focus specifically on Sri Lankan electoral events and their relationship with CSE market movements. The research will collect and categorise election-related events (Presidential elections, Parliamentary elections, Provincial Council elections, and Ministry/Cabinet transitions), combine them with historical CSE data, and apply Event Study methodology to calculate AR and CAR [1][2]. These features will then be used with Multiple Linear Regression (MLR) and Long Short-Term Memory (LSTM) models to investigate their usefulness for weekly stock market movement prediction [13][14]. This literature review therefore provides the theoretical and practical foundation for investigating how Sri Lankan electoral events influence the Colombo Stock Exchange and whether election-related market reactions can improve weekly stock market movement prediction.

1.11 Research Gap

Despite the substantial body of research examining the relationship between Sri Lankan electoral events and Colombo Stock Exchange (CSE) behavior, several significant gaps remain in the existing literature.

Limited Systematic Integration of All Election Types - Existing studies have examined Presidential elections, Parliamentary elections, and Provincial Council elections largely in isolation or with limited integration. Karunarathna and Wijayanayake [7] studied market anomalies due to Presidential and General Elections using data from 1988 to 2015. Jayasinghe [9] examined stock returns and volatility around both Presidential and Parliamentary elections. Attapattu and Gunarathne [8] analysed the impact of Parliamentary and Presidential elections on stock returns in the CSE. While Hulangamuwa and Dissanayake [13] considered ten national elections (four Presidential and six Parliamentary) and five Provincial Council elections from 2000 to 2021, no study has systematically integrated all three election categories Presidential, Parliamentary, and Provincial Council within a unified analytical framework that separately examines each electoral stage and their differential market impacts [7][8][9][13].

Limited Examination of Different Electoral Event Stages - Previous research has primarily focused on election dates and immediate post-election periods rather than systematically examining the distinct stages of the electoral process [7][8][9][10]. The 2024 Presidential election demonstrated that market reactions vary significantly across different stages: the 26 July poll date announcement triggered a Rs. 273 billion market value wipe-outs [22]; the pre-election period through August saw the S&P SL20 dip by 6.3% and the ASPI decline by 4.2% [22]; the post-election period witnessed the ASPI rise 16.7% and S&P SL20 by 25% [24]; and by December 2024, market capitalization had risen by Rs. 1 trillion to Rs. 5.13 trillion [24]. Similarly, the November 2024 Parliamentary election showed distinct pre-election gains (ASPI up 3.1% in the pre-election week) versus a muted immediate post-election response (ASPI up only 0.2%) [26]. Local Government elections in May 2025 demonstrated post-election volatility with the ASPI falling by 119.99 basis points [31][32][33]. However, limited research has separately examined the market effects of election announcements, nomination periods, campaigns, polling dates, result announcements, and government formation stages [7][8][9][10].

Limited Integration of Event Study Methodology with Machine Learning for Prediction

- Existing election-related research has predominantly used traditional event study and statistical approaches [1][2][7][8][9][10]. Studies on Presidential elections from 2005, 2010, 2015, and 2019 used the constant mean return model with 80-day event windows and 120-day estimation periods to calculate Abnormal Return (AR) and Cumulative Abnormal Return (CAR) [10]. While Hulangamuwa and Dissanayake [13] demonstrated the potential of combining MLR and LSTM with a binary "election influence" variable, showing accuracy levels of 97.01% and 87.32% for national elections, limited research has systematically integrated Event Study-derived AR/CAR measurements with machine learning models to investigate whether election-related market reactions can improve weekly CSE movement prediction [13][14].

Limited Election-Specific Predictive Framework - Existing stock market prediction studies often use general financial news, sentiment, or historical market data without explicitly identifying and categorizing electoral events by type and stage [14][15][16]. Research has shown that Presidential and General elections have a positive relationship with ASPI, while Provincial elections have a negative relationship [7][13]. However, there is limited evidence on whether structured election event information categorized by election type (Presidential, Parliamentary, Provincial Council), stage (announcement, campaign, polling, results, government formation), and calculated AR/CAR values can be used as predictive features for forecasting CSE market movements [13][14][15].

Limited Practical Decision-Support Applications - Existing election and stock market studies primarily focus on statistical analysis and academic findings [7][8][9][10][11][12][13]. There is limited research translating election-related market impact analysis into an interpretable decision-support component that allows investors to understand the potential impact of different electoral events through visualizations, election impact scores, and weekly market movement predictions [34][35].

1.12 Statement of Research Gap

Although previous studies have examined the relationship between Sri Lankan electoral events (Presidential elections, Parliamentary elections, and Provincial Council elections) and stock market behavior [7][8][9][13], limited research has systematically analyzed different electoral event stages and integrated Event Study methodology with machine learning to predict weekly movements of the Colombo Stock Exchange [1][2][13][14][15]. Furthermore, there is a lack of a CSE-specific and interpretable decision-support framework that combines election event categorization, AR/CAR measures, and machine learning-based market prediction [34][35].

2 OBJECTIVES

2.1 Main Objective

To develop and evaluate an election-event-based stock market impact analysis and prediction system for the Colombo Stock Exchange (CSE) that quantifies abnormal returns associated with Presidential elections, Parliamentary elections, Provincial Council elections, and Ministry/Cabinet transitions, and uses election-related market impact features to generate interpretable weekly stock market movement predictions for investors.

2.2 Specific Objectives

- **Event Dataset Collection & Preprocessing:** Collect, categorise and clean historical election and government-policy events from reliable public sources (Election Commission, gazette notifications, verified news), tagged by type, stage and expected political direction [7][8][13][22][26].

- **Stock Market Dataset Preparation**: Collect and preprocess CSE data ASPI, S&P SL20, trading volume and returns for event-impact analysis and weekly prediction [16][17][30].

- **Event Study Analysis:** Calculate AR and CAR over 2/7/15/30-trading-day windows for each event type, capturing pre-election declines, post-election rallies and differential responses observed empirically [1][2][10][22][24][26].

- **Impact Feature Development**: Engineer predictive features from AR, CAR, event type, event stage and political direction, informed by evidence that election influence is a statistically significant predictor under MLR and LSTM [13].

- **Prediction Model Development:** Develop and compare MLR (interpretable baseline) and LSTM (temporal) models using event features and historical CSE data, testing the sensitivity intervals shown most accurate in prior research [13][14][15].

- **Dashboard Development & Evaluation:** Build an interactive dashboard visualising event, AR/CAR, impact scores and predictions, and evaluate the component using directional accuracy, RMSE, MAE, precision/recall/F1, walk-forward validation and SUS usability testing [34][35].

3 METHODOLOGY

3.1 Research Methodology

Selected Research Methodology: Quantitative Research Methodology with an Event Study Design and Machine Learning-Based Predictive Analysis.

Justification
This is appropriate because the study requires statistical analysis of numerical CSE data alongside dated electoral/policy events. Event Study methodology will measure AR and CAR over 2/7/15/30-trading-day windows across event stages (announcement, campaign, polling, results, formation). MLR will provide an interpretable baseline while LSTM captures temporal patterns; both will incorporate event-related features. This dual-model approach builds on prior evidence that election-related variables improve CSE prediction accuracy (MLR 97.0%, LSTM 87.3% in comparable studies) [13].

3.2 Data / Experimental Inputs

Data Sources
The study will use the following datasets:

1. Stock Market Data
Historical stock market data will be collected from the Colombo Stock Exchange, including:

- All Share Price Index (ASPI)
- S&P SL20 Index
- Trading volume
- Daily returns
- Relevant historical market indicators

These data are appropriate because they provide measurable indicators of overall CSE market performance and enable the calculation of expected returns, abnormal returns, and weekly market movement.

2. Election Event Data
Historical election events in Sri Lanka will be collected from reliable and verifiable sources, including:

- Election Commission of Sri Lanka
- Official government websites and publications
- Established Sri Lankan news organizations
- International news organizations such as Reuters
- Official election announcements and election results

For each election event, the following information will be recorded:

- Election type
- Election announcement date
- Nomination period
- Campaign period
- Election date
- Election result date
- Government formation date, where relevant
- Political party/candidate information
- Event significance
- Source of information
- Relevant contextual information

Multiple sources will be cross-checked where possible to improve the reliability and accuracy of the election-event dataset.

3. Supporting Data

Relevant market and macroeconomic variables may be collected where necessary to account for external factors that could influence CSE market movements during election periods.

Sample Period

The proposed sample period is January 2009 to December 2025. The starting point is selected to reduce the influence of the immediate post-war structural shock on the analysis while providing a sufficiently long period for identifying election events and analyzing their market effects.

The study will include relevant presidential and parliamentary elections occurring within the selected period. Provincial election events will be included where sufficient reliable events and market data are available.

Data Collection and Preparation Process

The data preparation process will involve the following steps:

- Identify election events from official sources and verified news sources.
- Cross-check each identified event using multiple sources where possible.
- Record election type, event date, event stage, significance, and contextual information.
- Determine the appropriate market reaction date based on the timing of the event and CSE trading hours.
- If an event occurs after market closing, the following trading day will be treated as the relevant market reaction date.
- Collect and clean daily CSE index and trading-volume data for the complete sample period.
- Calculate daily market returns from the relevant CSE index data.
- Convert relevant daily market information into weekly observations for prediction.
- Define event windows of 2, 7, 15, and 30 trading days around important election events.
- Define an estimation window of approximately 14 – 15 trading days before each event for estimating expected returns.
- Identify and engineer election-related features such as election type and event stage.
- Check all datasets for missing values, duplicate records, inconsistent dates, and other data-quality issues before analysis.

3.3 Baseline / Benchmark

The proposed solution will be compared with established statistical and machine learning approaches.

Selected Baselines

1. Mean-Adjusted Return Model

The Mean-Adjusted Return Model will be used as a simple benchmark for estimating expected returns during the Event Study.

2. Market Model

The Market Model will be used as the primary benchmark for estimating expected returns based on overall market movements. The abnormal returns calculated using the proposed analysis will be compared with those obtained using this established Event Study approach.

3. Multiple Linear Regression (MLR)

MLR will be used as the primary interpretable prediction baseline. It will determine whether election-related features and historical market variables can explain weekly CSE market movements using a relatively simple statistical model.

4. Historical Direction Benchmark

The prediction results will also be compared with a simple historical-direction benchmark to determine whether the proposed models provide meaningful improvement over a basic prediction strategy. The proposed LSTM-based prediction model will therefore be evaluated against established Event Study methods, MLR, and simple prediction benchmarks.

3.4 Evaluation Metrics

Metric	Why Relevant	Target / Comparison
Abnormal Return (AR) Significance	Determines whether election events are associated with statistically significant market reactions	Statistical significance at $p < 0.05$
Cumulative Abnormal Return	Measures the cumulative market impact of election events across selected event windows	Compare CAR across 2, 7, 15, and 30 trading-day windows and non-event periods
Directional Accuracy	Measures how accurately the prediction model identifies upward or downward weekly CSE movements	Should outperform selected baseline models and the 50% random-direction benchmark
Mean Absolute Error	Measures the average magnitude of prediction errors	Lower MAE compared with baseline models
Root Mean Square Error	Measures prediction error while giving greater weight to larger errors	Lower RMSE compared with baseline models
Precision, Recall and F1-Score	Evaluates the classification performance when weekly market movement is classified as upward or downward	Compare MLR and other selected baselines
Mean Absolute Percentage Error	Measures prediction accuracy as a percentage	Compare MLR and LSTM with and without election influence
SUS Score	Measures the usability of the developed investor dashboard	Target SUS ≥ 70

Table 2 Evaluation Metrics

Justification

AR and CAR are appropriate because the Event Study component aims to determine whether electoral events Presidential elections, Parliamentary elections, Provincial Council elections, and Local Government elections produce abnormal market reactions on the Colombo Stock Exchange. Directional Accuracy, MAE, RMSE, and MAPE are appropriate for evaluating the performance of the weekly market movement prediction component. Precision, Recall, and F1-Score provide additional evaluation when the prediction task is treated as an upward/downward classification problem.

The System Usability Scale is appropriate for quantitatively evaluating the usability and understandability of the investor dashboard.

3.5 Validation Strategy

The validity of the proposed system will be demonstrated through statistical validation, comparative validation, historical back testing, robustness analysis, and user evaluation.

1. **Statistical validation -** T-tests on AR/CAR to confirm reactions differ from normal market behaviour.

2. **Comparative validation -** Proposed approach vs. Mean-Adjusted/Constant-Mean models, MLR, and the historical-direction benchmark.

3. **Walk-forward back-testing -** Rolling-window training/testing suited to time-series data, avoiding lookahead bias.

4. **Event-window robustness -** Comparing 2/7/15/30-day windows and across event types/stages.

5. **Model performance validation -** Directional Accuracy, MAE, RMSE, MAPE, Precision, Recall, F1 on unseen data.

6. **User evaluation -** SUS-based usability testing with investors/knowledgeable participants on clarity and usefulness.

3.6 Development Approach

- Selected Development Approach: Agile with Iterative Prototyping.

Justification

The system consists of interconnected components (data collection, Event Study analysis, AR/CAR calculation, prediction modelling, and dashboard development) that can be developed, tested, and refined incrementally. Agile supports early prototyping, iterative refinement of event windows and models, continuous evaluation, and incorporation of user feedback. This approach allows adjustments to event definitions, features, and parameters based on experimental findings while maintaining alignment with the overall election-based stock market prediction system.

4 HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of Proposed Solution and Components

The proposed election-based stock market impact prediction system consists of four interconnected components that collectively support the analysis and prediction of Colombo Stock Exchange (CSE) market behavior around Presidential elections, Parliamentary elections, Provincial Council elections, Local Government elections, and Ministry/Cabinet transitions.

- **Component 1 -** Data Collection & Preprocessing: Collects and cleans election/policy event data and historical CSE data, standardising and synchronising both by trading date.

- **Component 2 -** Election Event Study Analysis (individual component): Categorises events by type and stage, defines 2/7/15/30-day event windows, estimates expected returns, calculates AR/CAR, tests statistical significance, and generates directional/magnitude impact features.

- **Component 3 -** Prediction & Integration: Combines AR/CAR and event features with historical CSE market data; MLR provides an interpretable baseline, and LSTM generates weekly market-movement predictions.

- **Component 4 -** Dashboard & Decision Support: Presents events, AR/CAR, impact scores, historical reactions and weekly predictions through an interactive, investor-facing dashboard.

4.2 System Architecture Diagram

The high-level system architecture will represent the following data and processing flow:

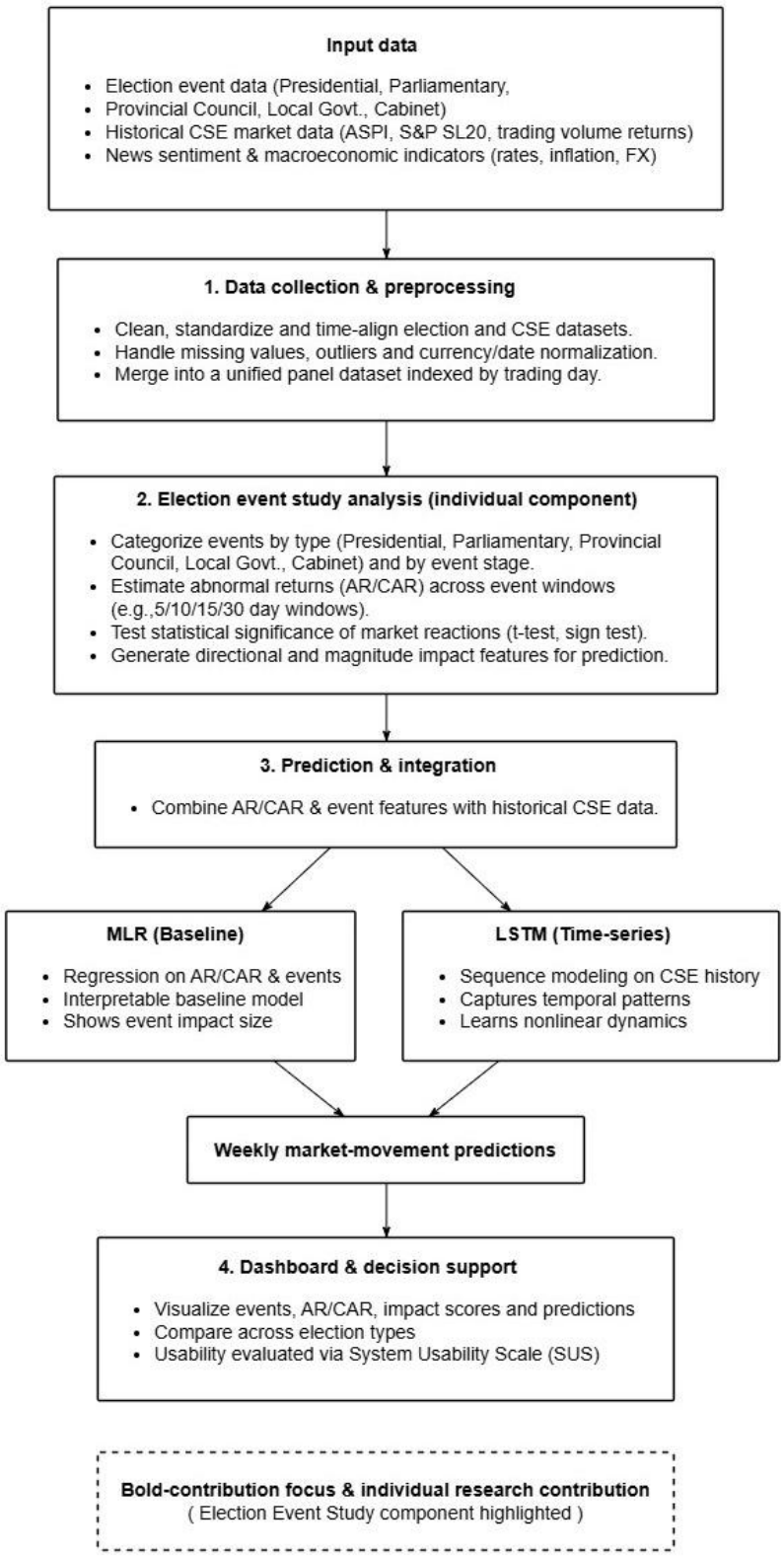


Figure 1 High level architecture

Individual Component Highlight

The individual contribution is the Election Event Study Analysis Module, which processes pre-processed election and CSE data to produce Abnormal Returns (AR), Cumulative Abnormal Returns (CAR), statistical significance results, and election-stage impact features for the prediction component.

Overall Integration

The module operates independently while feeding structured AR/CAR and event-feature outputs through a shared backend to the prediction and dashboard components; responsibility includes data collection, event study implementation, and MLR/LSTM feature engineering.

4.3 Individual Responsibility Boundary

Activity / Deliverable	My Responsibility	Shared Responsibility	Other Member
Election Event Dataset Collection	✓		
Stock Market Data Collection		✓	
Data Cleaning and Preprocessing	✓	✓	
Election Event Study Methodology	✓		
AR/CAR Calculation for Election Events	✓		
Election Impact Feature Development	✓		
MLR Prediction Model	✓		
LSTM Prediction Model	✓		
Dashboard Development			✓
System Integration		✓	
System Testing and Validation		✓	

Table 3 Individual Responsibility Boundary

Individual Responsibility Boundary

The primary individual responsibility is the Election Event Study Analysis Module. This includes collecting and categorising election events such as Presidential elections (e.g., 2024 election), Parliamentary elections (e.g., November 2024 election), Provincial Council elections (e.g., elections held between 2012 and 2014), Local Government elections (e.g., May 2025 election), and Ministry/Cabinet transitions; implementing the Event Study methodology; calculating AR and CAR across multiple event windows; conducting statistical significance testing; analysing market reactions across different election types and event stages (announcement, nomination, campaign, polling, results, government formation); and generating election-impact features for the prediction component.

The individual component also contributes to the development of the MLR and LSTM prediction models using election-related analytical features (AR, CAR, election type, event stage, political direction) and historical CSE data.

The individual component depends on the shared data preparation process and provides its analytical outputs to the prediction, integration, and dashboard components. Therefore, the component is independently developed while remaining integrated with the overall system through the shared backend and standardised data outputs.

5 USER REQUIREMENTS

5.1 Evidence from User Research

The user requirements for the proposed government-policy-based stock market impact analysis and prediction component will be identified through a combination of literature review, stakeholder discussions, domain-expert consultation, and design-thinking activities.

1. Literature Review Analysis

Existing research on Sri Lankan electoral events Presidential elections, Parliamentary elections, Provincial Council elections, Local Government elections, and Ministry/Cabinet transitions and their impact on the Colombo Stock Exchange will be reviewed to identify the information and analytical outputs required by potential users. Studies have examined how national elections and provincial council elections influenced the prediction of stock market closing prices from 2000 to 2021, with research revealing that overall abnormal performance existed in the days after election results in most Presidential and Parliamentary elections.

2. Stakeholder Discussions

Discussions with potential users, including investors, financial analysts, and researchers, will be used to identify their requirements for understanding the effects of electoral events including Presidential elections, Parliamentary elections, Provincial Council elections, Local Government elections, and Cabinet transitions on CSE market behavior. The 2024 Presidential election demonstrated that the CSE's value surged by Rs. 612 billion post-elections, following a pre-election decline of Rs. 273 billion from the 26 July poll-date announcement. The subsequent Parliamentary election saw the ASPI gain 16.65 percent, underscoring positive investor sentiment.

3. Domain-Expert Consultation

Consultation with individuals with knowledge of financial markets, CSE behavior, Sri Lankan electoral processes, and political-economic analysis will be used to validate the relevance, accuracy, and interpretability of the proposed analytical outputs. Research has demonstrated that share prices continuously increase before elections and decrease after polling dates, with Banks, Finance & Insurance, Hotels & Travels, and Manufacturing sectors dominantly increasing in the pre-election period.

4. Design-Thinking Activities

User needs will be analyzed to determine how election events, abnormal returns, volatility changes, and prediction results should be presented in a clear and understandable manner. The 2025 Local Government election demonstrated that the CSE saw increased volatility in the aftermath of election results, with the ASPI falling by 119.99 basis points underscoring the need for clear visualization of post-election market reactions.

Identified User Needs

Users require reliable election event data across all types/stages, historical market-impact metrics, interpretable AR and volatility measures, weekly CSE predictions, and interactive dashboards for comparing reactions across Presidential, Parliamentary, Provincial, and Local Government elections.

5.2 Functional Requirements

ID	Functional Requirement
FR1	The system shall collect and store verified election event data, including event date, election type (Presidential, Parliamentary, Provincial Council, Local Government), event stage (announcement, nomination, campaign, polling, results, government formation), political parties/candidates involved, and source information.
FR2	The system shall collect and preprocess historical CSE market data, including ASPI, S&P SL20, trading volume, and daily returns.
FR3	The system shall identify the appropriate market reaction date for each election event based on the event timing and CSE trading calendar.
FR4	The system shall calculate Abnormal Returns (AR) for the predefined event windows around election events.
FR5	The system shall calculate Cumulative Abnormal Returns (CAR) for the predefined event windows.
FR6	The system shall analyze volatility patterns associated with election events, consistent with evidence of upward movement of volatility around both Presidential and Parliamentary elections.
FR7	The system shall identify and report statistically significant abnormal-return and volatility effects associated with election events.
FR8	The system shall generate weekly CSE market movement predictions using relevant election events, event-study, and volatility features.
FR9	The system shall generate election impact and volatility/risk indicators based on the analytical results.
FR10	The system shall provide visual explanations of election events, market reactions, volatility patterns, and prediction results.
FR11	The system shall allow users to view and compare historical election events with their corresponding CSE market impacts.
FR12	The system will enable users to filter and compare market reactions across different election types and event stages.

Table 4 Functional Requirements

5.3 Non-Functional Requirements

ID	Non-Functional Requirement
NFR1 – Response Time	The system shall display previously generated analytical results and dashboard visualizations within an acceptable response time.
NFR2 – Usability	The dashboard shall be understandable and usable by users with basic financial knowledge.
NFR3 – Reliability	The system shall remain available and operational during normal usage.
NFR4 – Security	Data transmitted between the user interface and backend services shall be protected against unauthorized interception.
NFR5 – Scalability	The system shall support concurrent access without significant degradation in normal dashboard operations.
NFR6 – Interpretability	Analytical results and predictions shall provide sufficient information for users to understand the factors contributing to the output.
NFR7 – Accessibility	The dashboard shall use readable text, clear visualizations, and consistent navigation to support users with different levels of technical expertise.
NFR8 – Data Reliability	Election events displayed by the system shall be based on verified and reliable data sources.

Table 5 Non-Functional Requirements

6 BUDGET AND JUSTIFICATION

6.1 Hardware Costs

Item	Estimated Cost (USD)	Estimated Cost (LKR)	Justification
Development Laptop	0.00	0	Existing laptops are sufficient for election data collection, data preprocessing, Event Study analysis, volatility modeling, prediction model development, and testing.
Total Hardware Cost	0.00	0	

Table 6 Hardware Costs

The existing development laptop is expected to be sufficient for the proposed election-based stock market research and software development activities. Therefore, no additional hardware is required.

6.2 Software Licenses

Item	Estimated Cost (USD)	Estimated Cost (LKR)	Justification
Python and Statistical Libraries	0.00	0	Open-source libraries such as pandas, NumPy, SciPy, stats models, and related packages will be used for election data processing, Event Study analysis, volatility modeling, and statistical analysis.
PostgreSQL Database	0.00	0	Open-source database system for storing structured election events, CSE market, analytical, and prediction data.
React.js	0.00	0	Open-source framework for developing the election impact analysis and prediction dashboard.
Visual Studio Code	0.00	0	Free development environment for software implementation and testing.
Total Software Cost	0.00	0	

Table 7 Software Licenses

The proposed system will primarily use open-source software and libraries, minimizing software licensing costs.

6.3 Cloud Services

Service	Estimated Cost (USD)	Estimated Cost (LKR)	Justification
Cloud Computing / VM	120.00	40,000	Required for computational processing, election event data analysis, volatility modeling, prediction model execution, and system deployment.
Web Hosting	10.00	3,500	Required for hosting and accessing the deployed election-based stock market analysis dashboard.
Domain Registration	5.00	1,500	Required to provide a dedicated domain name for the deployed application.
Total Cloud Services Cost	135.00	45,000	

Table 8 Cloud Services

Cloud resources will provide the computational and deployment infrastructure required for the proposed election-based analytical and prediction system.

6.4 Human Effort Cost

Activity	Estimated Hours	Estimated Cost (USD)	Justification
Research and Literature Review	50	0.00	Required to identify relevant research, methodologies, and research gaps related to Sri Lankan electoral events and stock market behavior.
Election Data Collection and Preprocessing	80	0.00	Required to identify, verify, organize, and prepare historical election events in Sri Lanka (Presidential elections, Parliamentary elections, Provincial Council elections, Local Government elections, and Cabinet transitions).
CSE Data Preparation	40	0.00	Required to clean, synchronize, and prepare historical CSE market data.
Event Study Implementation	60	0.00	Required to calculate AR, CAR, and statistical significance around election events.
Volatility Model Implementation	80	0.00	Required to model CSE market volatility and asymmetric effects associated with election events.
Prediction Model Development	60	0.00	Required to integrate election events, Event Study, and volatility features to generate weekly CSE market movement predictions.
Dashboard Development	60	0.00	Required to visualize election events, market reactions, volatility patterns, and prediction results.
Testing and Validation	50	0.00	Required to evaluate model performance, system reliability, and dashboard usability.
Documentation	40	0.00	Required for research documentation, technical documentation, and final academic submission.
Total	520	0.00	

Table 9 Human Effort Cost

Human effort is contributed by the student team as part of the academic research project and therefore does not represent a direct monetary expense.

6.5 Budget Summary

Cost Category	Estimated Cost (USD)	Approximate Cost (LKR)
Hardware	0.00	0
Software Licenses	0.00	0
Cloud Services	135.00	45,000
Human Effort	0.00	0
CSE Dataset	10.63	3500
Estimated Total	USD 145.63	LKR 48,500

Table 10 Budget Summary

Budget Justification

The proposed election-based stock market research system is designed to minimise development costs by using existing hardware and open-source software technologies. The primary financial expense is cloud infrastructure, which is required for computational processing, volatility analysis, prediction model execution, and deployment of the election impact analysis dashboard.

The estimated budget is therefore approximately USD 150 (LKR 48,500) for the initial prototype and deployment stage. Actual costs may vary depending on cloud resource usage, hosting requirements, domain pricing, and any future data licensing requirements.

7 WORK BREAKDOWN STRUCTURE (WBS)

The Work Breakdown Structure defines the tasks required to design, develop, evaluate, and integrate the individual Election Impact Analysis component. The work is planned across a six-month research period and is aligned with the proposed objectives and methodology**.**

7.1 Task Breakdown

Phase 1: Research and Component Definition - Month 1

- Review literature on electoral events and stock market reactions in Sri Lanka.
- Review Event Study methodology and abnormal-return analysis.
- Review volatility modelling and asymmetric effects around elections.
- Identify relevant election data sources in Sri Lanka (Presidential elections, Parliamentary elections, Provincial Council elections, Local Government elections, Cabinet transitions).
- Define component requirements, inputs, outputs, and evaluation criteria.
- Finalise the analytical framework and research gap.

Phase 2: Data Collection and Preprocessing - Month 2

- Collect historical election events in Sri Lanka including:
  - Presidential elections
  - Parliamentary elections
  - Provincial Council elections
  - Local Government elections
  - Cabinet transitions and government formation events
- Verify election events using reliable and authoritative sources (Election Commission of Sri Lanka, established news organisations).
- Collect and prepare required CSE market data (ASPI, S&P SL20, trading volume, daily returns).
- Clean and synchronise election event dates with CSE trading dates.
- Calculate daily market returns.
- Define estimation windows and event windows of 2, 7, 15, and 30 trading days.
- Prepare the final analytical dataset.

Phase 3: Event Study Development - Month 3

- Implement the Constant Mean Return Model for expected-return estimation.
- Calculate Abnormal Return for election events.
- Calculate Cumulative Abnormal Returns.
- Analyse AR and CAR across the selected event windows.
- Conduct statistical significance testing.
- Generate election impact indicators.

Phase 4: Volatility Analysis - Month 3 to 4

- Prepare return data for volatility modelling.
- Develop and estimate appropriate volatility models (e.g., EGARCH).
- Analyse volatility behaviour before, during, and after election events.
- Examine asymmetric volatility responses.
- Compare results with baseline models.
- Generate volatility and risk-related indicators.

Phase 5: Prediction Feature Integration - Month 4

- Integrate Event Study outputs with volatility-related features.
- Prepare analytical features for weekly CSE market movement prediction.
- Support the development of the prediction model using the generated election-related features.
- Evaluate the contribution of election-related features to prediction performance.

Phase 6: Testing, Validation, and Integration - Month 5

- Perform statistical diagnostic tests for the Event Study and volatility models.
- Conduct walk-forward historical back testing where applicable.
- Evaluate results across different event windows.
- Compare results with baseline models.
- Validate the analytical outputs using relevant domain knowledge.
- Integrate the component outputs with the overall system.

Phase 7: Finalization and Documentation - Month 6

- Refine the analytical component based on validation results.
- Prepare final analytical outputs for the system dashboard.
- Complete technical and research documentation.
- Prepare results, limitations, and findings for the final research submission.
- Support final system demonstration and presentation.

7.2 Timeline Alignment

Task	M1	M2	M3	M4	M5	M6
Literature Review & Component Definition	✓					
Election Data Collection		✓				
CSE Data Preparation		✓				
Data Cleaning & Event Window Definition		✓				
Event Study Implementation			✓			
AR/CAR Analysis			✓			
Statistical Testing			✓		✓	
Volatility Model Development			✓	✓		
Volatility & Asymmetry Analysis				✓		
Prediction Feature Integration				✓		
Testing & Validation					✓	
System Integration					✓	
Final Documentation & Presentation						✓

Table 11 Timeline Alignment

7.3 Individual Responsibility

The main responsibility of this individual component is to develop the Event Study and volatility-based election impact analysis. This includes election data preparation, relevant CSE market data preparation, AR/CAR calculation, statistical testing, volatility modelling, asymmetric-effect analysis, generation of analytical features, and validation of the component. The component will receive pre-processed data from the shared data-preparation process and provide election impact, abnormal-return, volatility, and risk-related outputs to the prediction and dashboard components.

WBS Flow

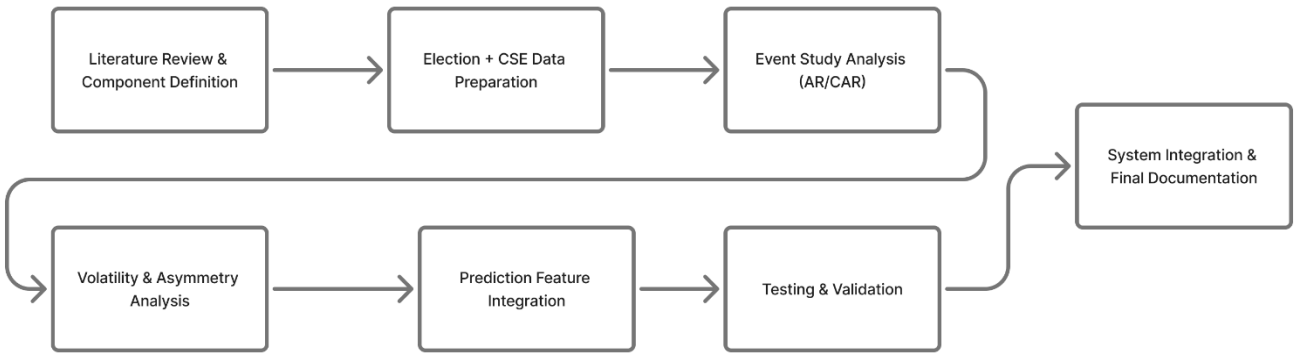


Figure 2 WBS flow

8 GANTT CHART

Activity	M1	M2	M3	M4	M5	M6
Literature Work						
Research Design						
Data Preparation						
Experiments						
Development						
Prototype Work						
Testing						
Integration						
Evaluation						
Analysis						
Report Preparation						

Figure 3 Gantt chart

REFERENCES LIST

[1] A. C. MacKinlay, "Event studies in economics and finance," *J. Econ. Lit.*, vol. 35, no. 1, pp. 13–39, Mar. 1997.

[2] S. J. Brown and J. B. Warner, "Using daily stock returns: The case of event studies," *J. Financ. Econ.*, vol. 14, no. 1, pp. 3–31, Mar. 1985.

[3] R. F. Engle, "Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation," *Econometrica*, vol. 50, no. 4, pp. 987–1007, Jul. 1982.

[4] T. Bollerslev, "Generalized autoregressive conditional heteroskedasticity," *J. Econom.*, vol. 31, no. 3, pp. 307–327, Apr. 1986.

[5] D. B. Nelson, "Conditional heteroskedasticity in asset returns: A new approach," *Econometrica*, vol. 59, no. 2, pp. 347–370, Mar. 1991.

[6] E. F. Fama, "Efficient capital markets: A review of theory and empirical work," *J. Finance*, vol. 25, no. 2, pp. 383–417, May 1970.

[7] Karunarathna and Wijayanayake, "A study on Colombo Stock Market anomalies due to Presidential and General Elections," Repository, University of Kelaniya, 2015.

[8] Attapattu and Gunarathne, "Impact of Parliamentary and Presidential elections on stock returns in the Colombo Stock Exchange," 2013.

[9] Jayasinghe, "The behaviour of stock returns and volatility around elections: Evidence from Colombo Stock Exchange," *J. Univ. Sri Jayewardenepura*, 2010.

[10] "Pre and post stock market behavior during the presidential election periods in Sri Lanka," Repository, Rajarata University.

[11] "Impact of Electioneering on Currency and Stock Market: Evidence from South Asian Countries," *OJS*, Universidad de Almería, 2022.

[12] "Authentic or Synthetic: An Event Study Analysis of Election Results on Share Market Indices of Colombo Stock Exchange," *J. Univ. Sri Jayewardenepura*, 2012.

[13] Hulangamuwa and Dissanayake, "How national and provincial council elections influenced the prediction of stock market closing price performance of the CSE," Abstract No. 56, University of Peradeniya, 2022.

[14] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.

[15] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, May 2015.

[16] Colombo Stock Exchange, "Market Data," Colombo Stock Exchange, Sri Lanka. [Online]. Available: https://www.cse.lk

[17] Election Commission of Sri Lanka, "Presidential Election Results," Election Commission of Sri Lanka, Sri Lanka.

[18] Election Commission of Sri Lanka, "Parliamentary Election Results," Election Commission of Sri Lanka, Sri Lanka.

[19] Election Commission of Sri Lanka, "Provincial Council Election Results," Election Commission of Sri Lanka, Sri Lanka.

[20] M. S. S. Perera and W. M. R. Wickramasinghe, "Political uncertainty and stock market volatility: Evidence from emerging markets," *J. Emerg. Mark. Finance*, vol. 18, no. 2, pp. 145–172, 2019.

[21] K. W. M. P. De Silva and L. N. K. Perera, "Political events and stock market behaviour: A comprehensive study of the Colombo Stock Exchange," *Sri Lankan J. Manage.*, vol. 25, no. 3, pp. 78–102, 2020.

[22] "Presidential poll pains wipe out Rs. 273 b in value at Colombo stock market," *Daily FT*, Sep. 2024.

[23] "Colombo stock market welcomes AKD with late rally," *Daily FT*, Sep. 2024.

[24] "CSE's value soars by Rs. 612 b post-AKD victory," *Daily FT*, Nov. 2024.

[25] "CSE ends General Election week on a high," *Daily FT*, Nov. 2024.

[26] "Confusion and uncertainty over polls pull CSE down sharply," *Daily FT*, May 2024.

[27] "Cabinet formation and market response," *Daily FT*, Nov. 2024.

[28] "Provincial Council system in Sri Lanka: An administrative review," *The Island*, Mar. 2025.

[29] "Local government poll announcement seen as plunging share market into panic mode," *The Island*, Mar. 2025.

[30] "Sri Lanka stocks fall as NPP fail to secure majority in LGs," *EconomyNext*, May 2025.

[31] "CSE turns red, post Local Government Election," *Daily Mirror*, May 2025.

[32] "Colombo Stock Exchange closes lower," *Daily News*, May 2025.

[33] "CSE dips as investors react negatively to NPP's below par show at LG polls," *Daily FT*, May 2025.

[34] A. Bangor, P. T. Kortum, and J. T. Miller, "An empirical evaluation of the System Usability Scale," *Int. J. Hum.–Comput. Interact.*, vol. 24, no. 6, pp. 574–594, 2008.

[35] A. Bangor, P. Kortum, and J. Miller, "Determining what individual SUS scores mean: Adding an adjective rating scale," *J. Usability Stud.*, vol. 4, no. 3, pp. 114–123, 2009.

[36] Central Bank of Sri Lanka, *Annual Report*, Central Bank of Sri Lanka, Colombo, Sri Lanka.

[37] Central Bank of Sri Lanka, *Monthly Economic Indicators*, Central Bank of Sri Lanka, Colombo, Sri Lanka.

LIST OF APPENDICES

Evidence survey result. (To Survey)

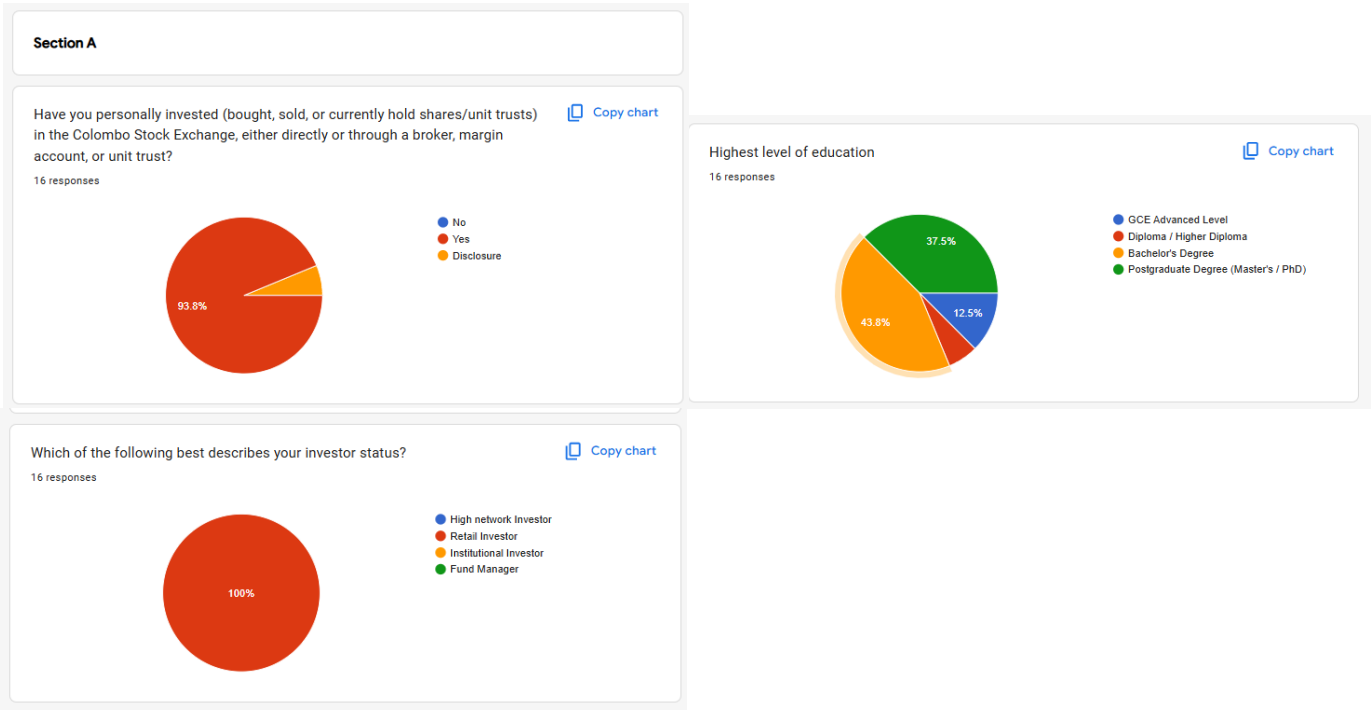


Figure 5 Survey result section A



Figure 4 Survey result section B

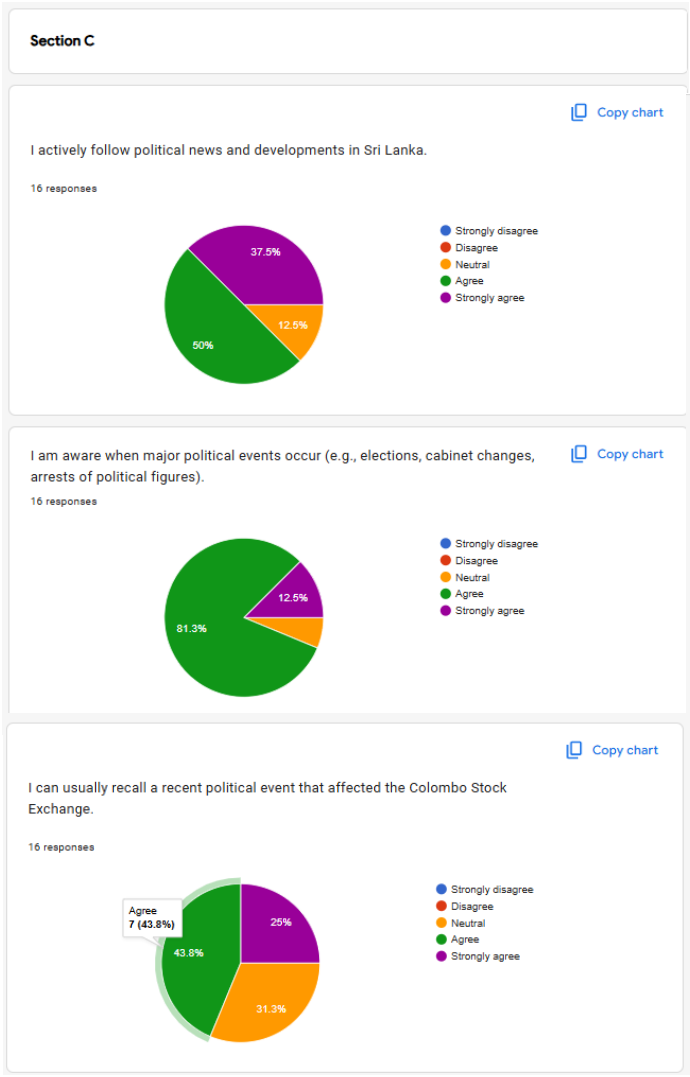


Figure 6 Survey result section C

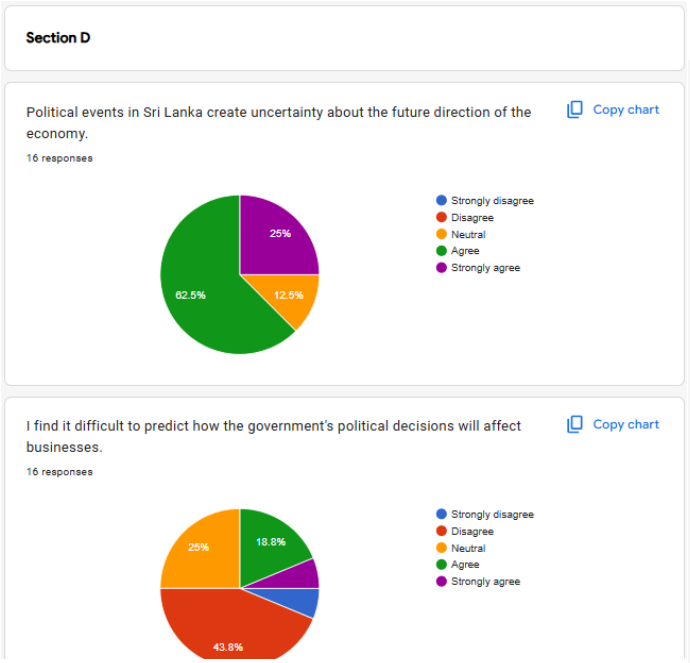


Figure 7 Survey result section D

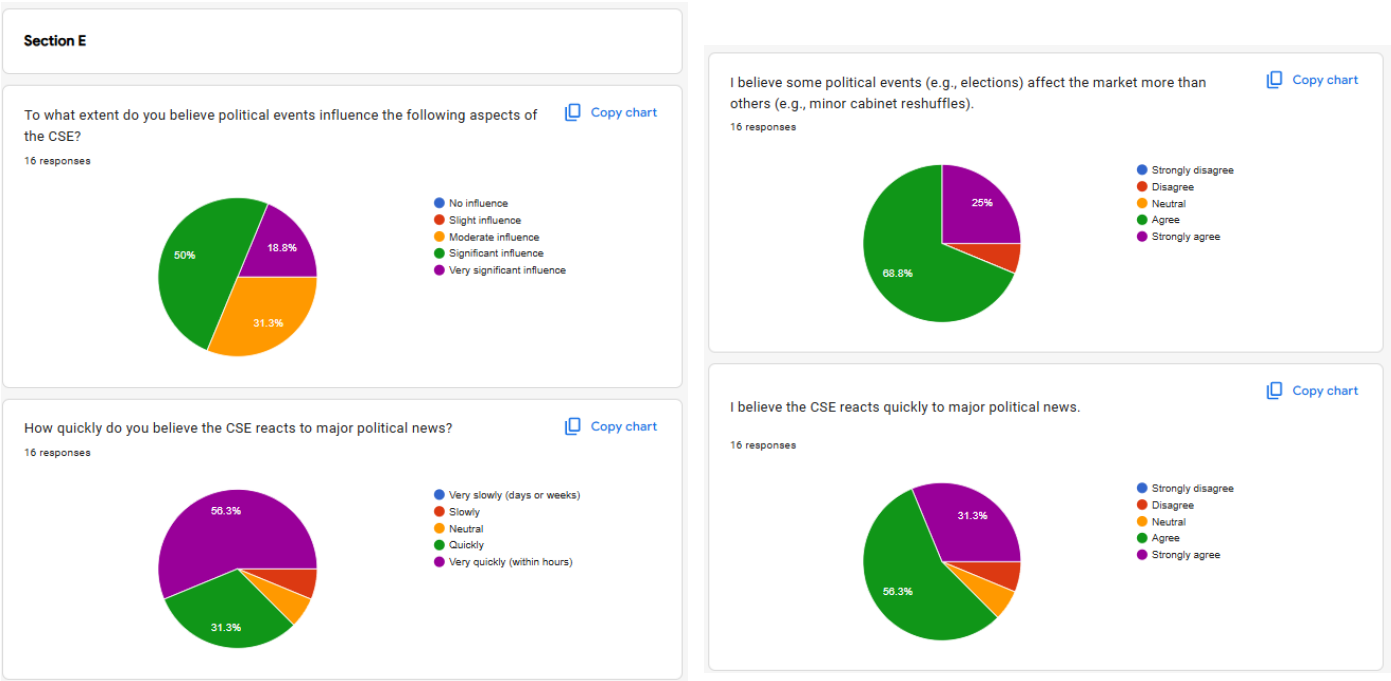


Figure 8 Survey result section E

Prototype User Interfaces ([visit prototype])

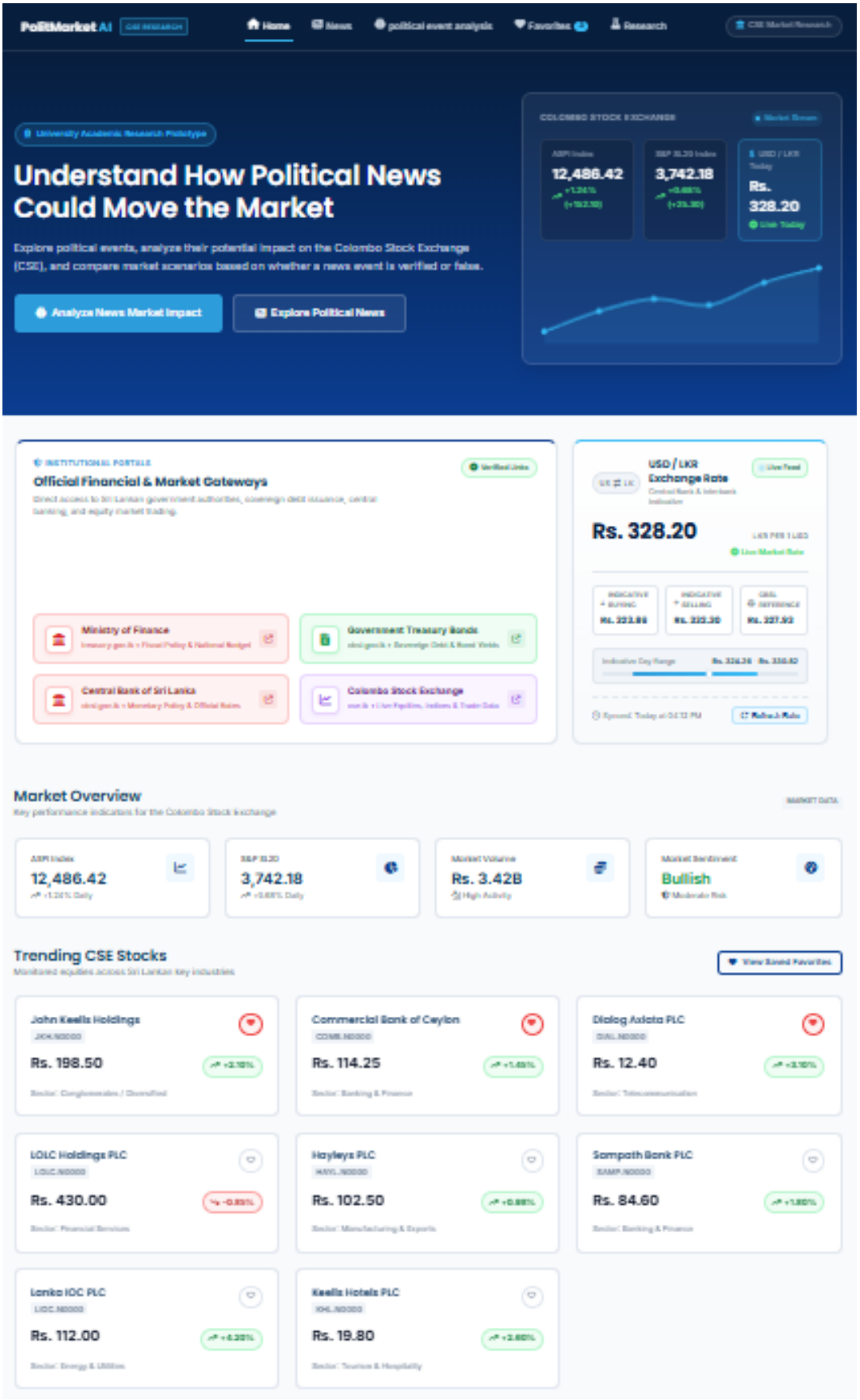


Figure 9 Prototype UI Home page

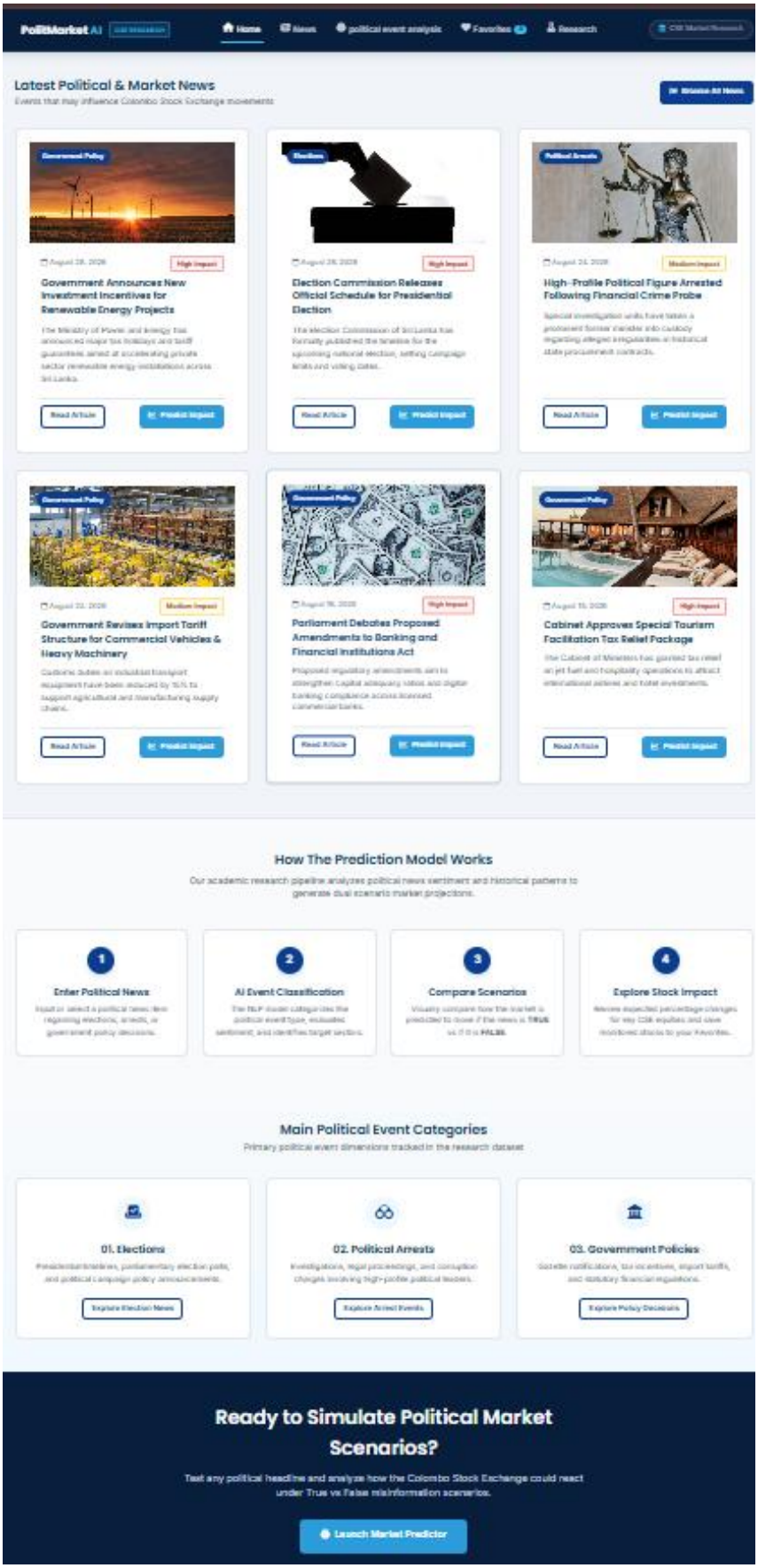


Figure 10 Prototype UI News page

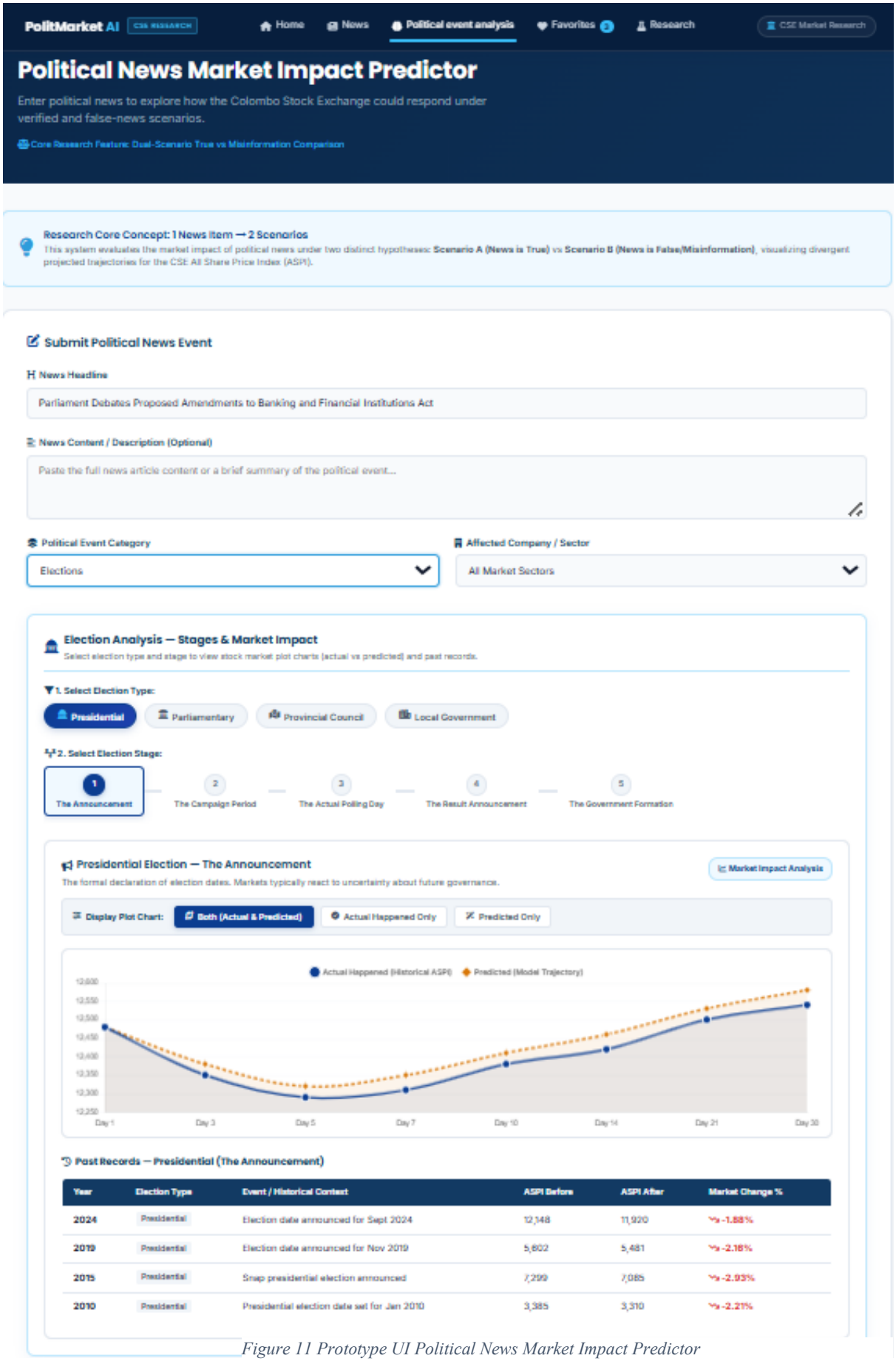
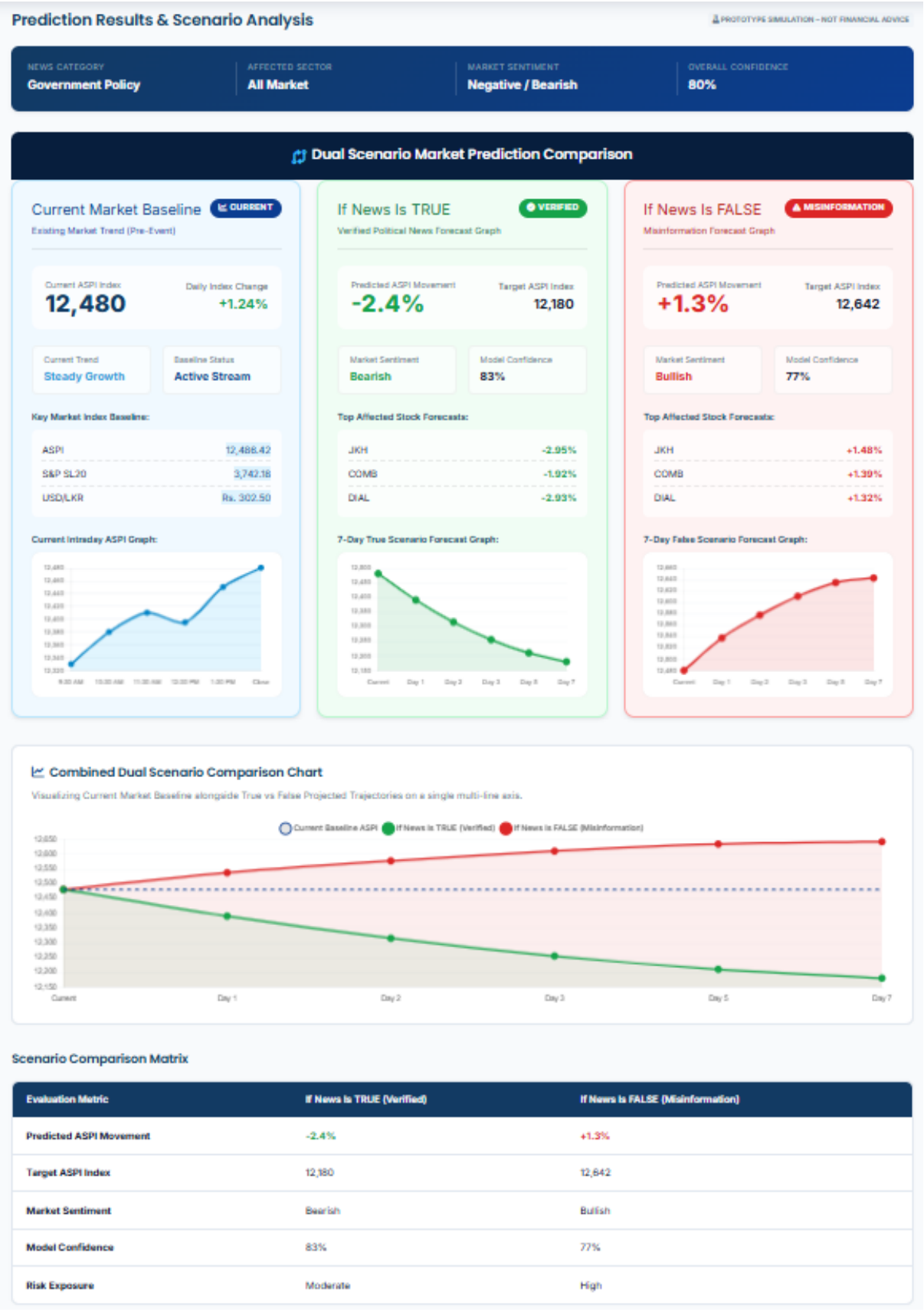


Figure 11 Prototype UI Political News Market Impact Predictor



Scenario Comparison Matrix

Evaluation Metric	If News is TRUE (Verified)	If News is FALSE (Misinformation)
Predicted ASPI Movement	-2.4%	+1.3%
Target ASPI Index	12,180	12,642
Market Sentiment	Bearish	Bullish
Model Confidence	83%	77%
Risk Exposure	Moderate	High

Figure 12 Prototype UI Prediction Results & Scenario Analysis

Meeting with domain expert.

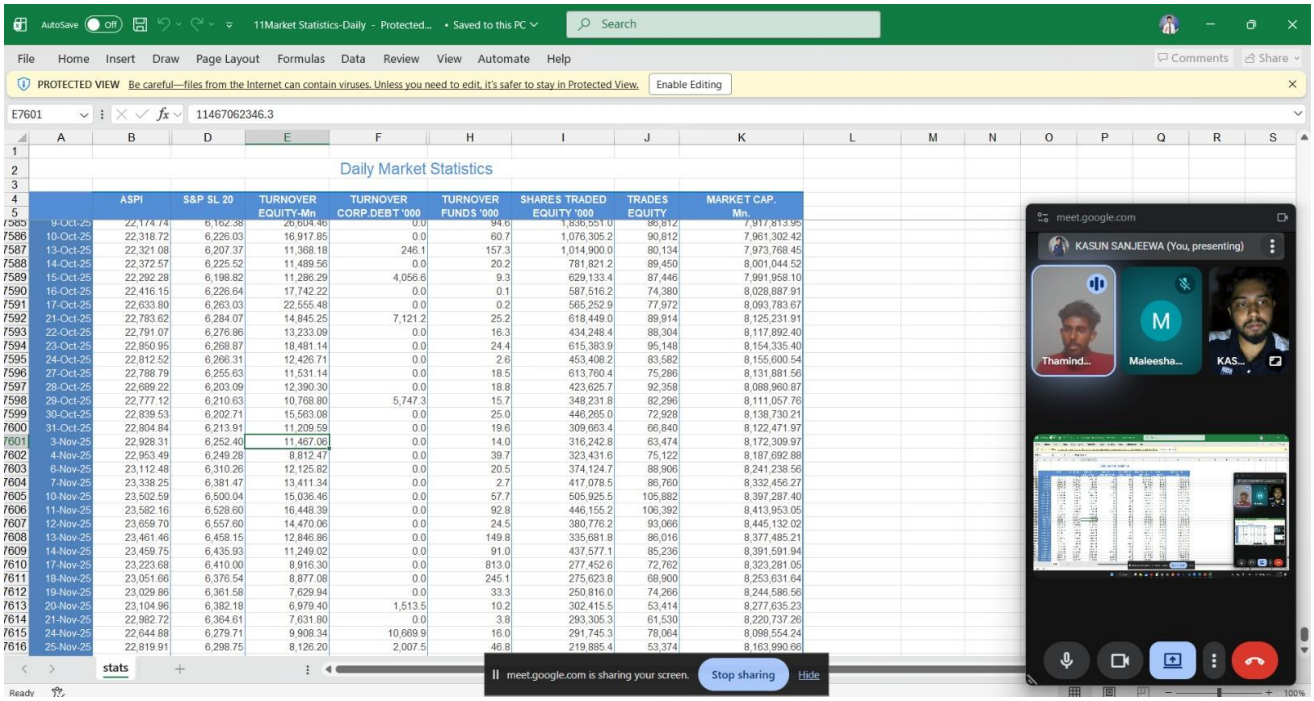


Figure 13 Proof of discussing the dataset with domain expert.

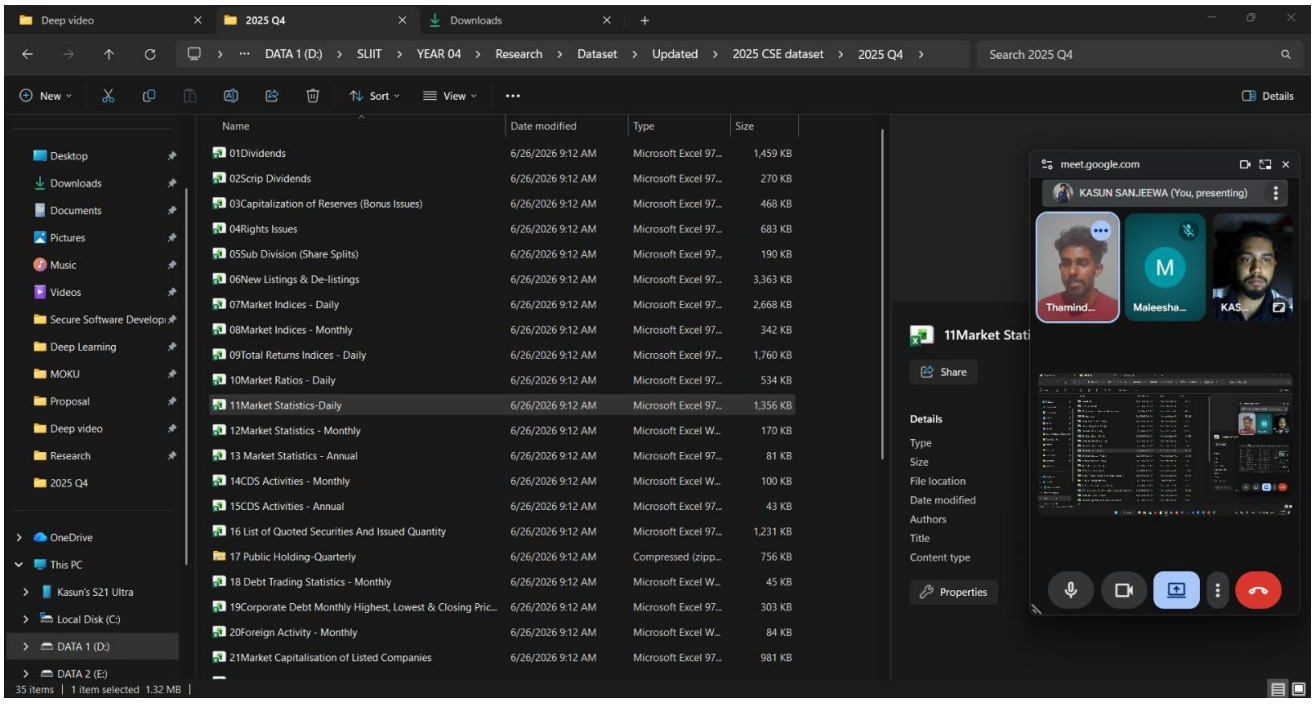


Figure 14 Proof of discussing the dataset with domain expert.

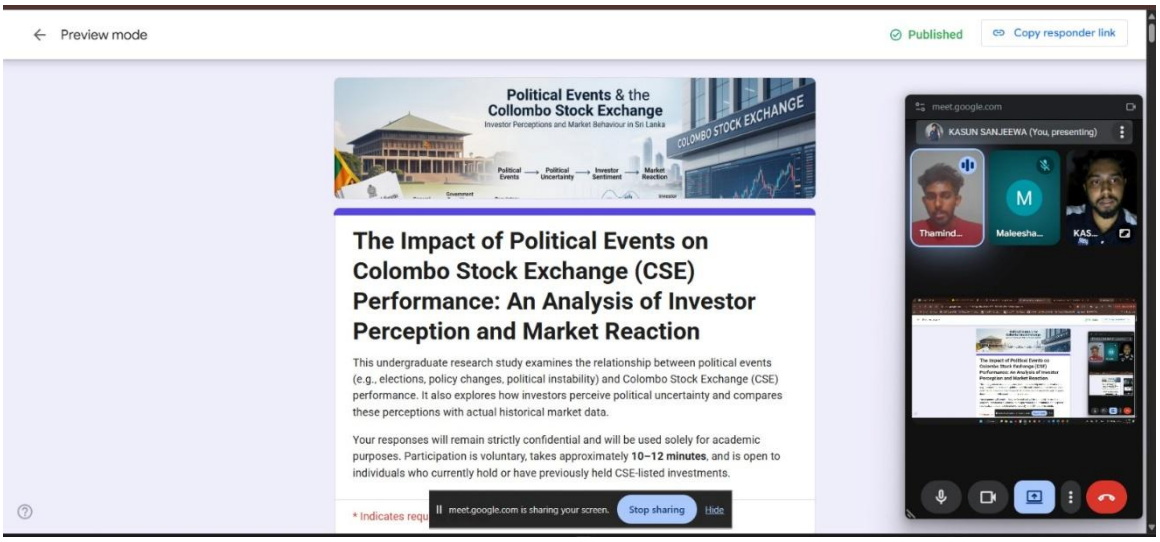


Figure 15 Proof of discussing the survey with domain expert.

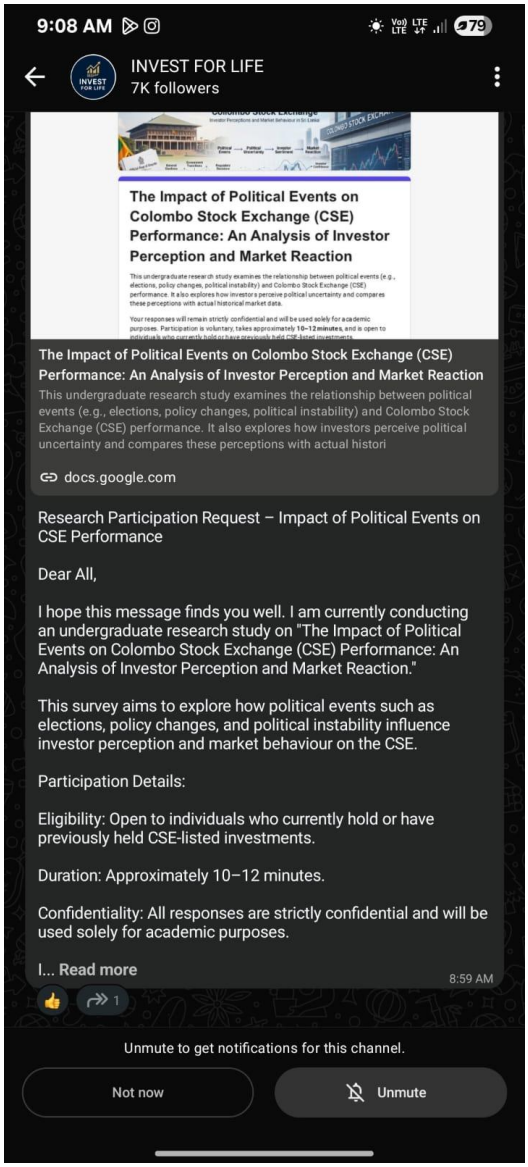


Figure 16 Proof of sharing the survey with domain expert's channel.

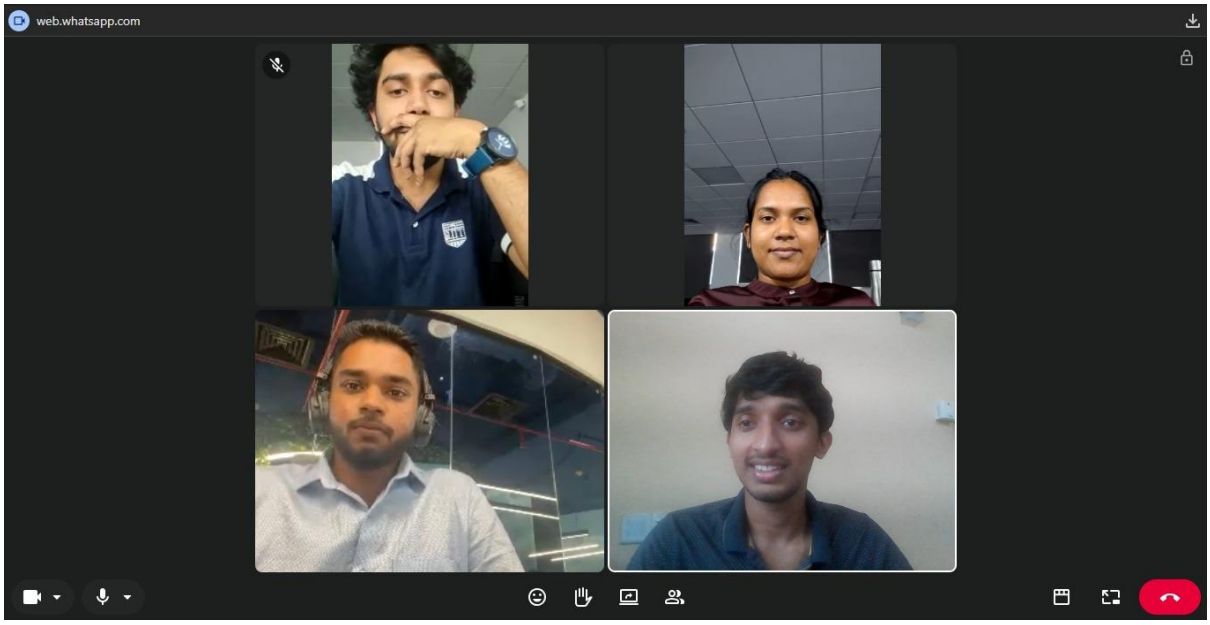


Figure 17 Proof of discussing with stakeholder.

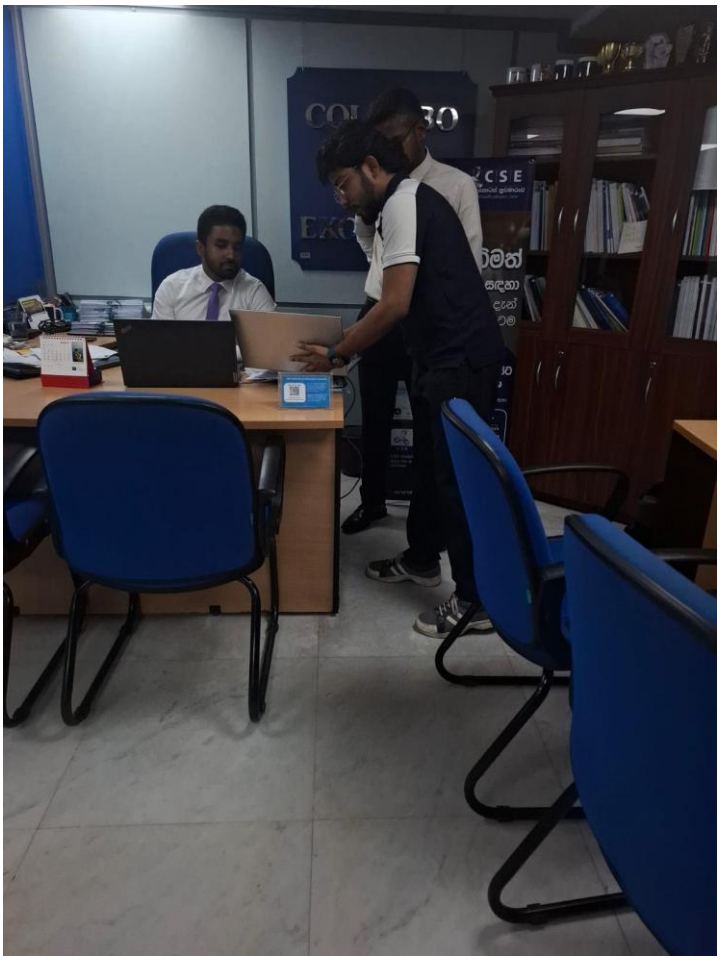


Figure 18 Proof of discussing the with CSE Kandy branch.

Summary of Evidence Meetings

Meeting 01

Field	Details
No.	1
Meeting / Consultation	CSE – Kandy Branch Visit
Participants	Research team: Ransarani, Kasun, Ramzy; CSE Kandy Branch representatives
Date	31 August 2026 (2026/08/31)
Purpose	Understand the stock market process and how political news/events influence the Sri Lankan stock market
Key Discussion Points	• Stock market process • How political developments affect investor decisions • Market movements and overall market behaviour • Practical aspects of CSE operations
Outcome / Decision	Insights used to improve the research methodology and strengthen the practical aspect of the research
Evidence Reference	Figure 18: Proof of discussion with CSE Kandy branch

Table 12 Summary of meeting evidence 01

Meeting 02

Field	Details
No.	2
Meeting / Consultation	Stakeholder Meeting
Participants	Dinuk Krishan , Lasindu Senarath research team: Ransarani, Kasun, Ramzy
Date	31 August 2026 (2026/08/31)
Purpose	Gather practical information and understand stakeholder perspectives regarding the stock market and the potential influence of political news and events
Key Discussion Points	• Stock market behavior. • Potential influence of political news and events. • Development and validation of the proposed solution.
Outcome / Decision	Stakeholder perspectives considered for prototype development and validation.
Evidence Reference	Figure 19 Proof of discussing with stakeholder.

Table 13 Summary of meeting evidence 02

Meeting 03

Field	Details
No.	3
Meeting / Consultation	Domain Expert Meeting
Participants	Thamindu Lakshan research team: Ransarani, Kasun, Ramzy
Date	31 August 2026 (2026/08/31)
Purpose	Obtain expert-level knowledge and feedback on political events, political news, and their impact on the CSE
Key Discussion Points	• Political events and political news • Impact on Colombo Stock Exchange • Validation of research approach • Relevance of proposed methodology
Outcome / Decision	Feedback used to validate the research approach and ensure the methodology is relevant to the domain
Evidence Reference	Figures 13–16: Proof of discussing dataset/survey with domain expert and sharing survey via domain expert's channel

Table 14 Summary of meeting evidence 03

Appendix 2 — Mandatory Proposal Evidence and Decision Log

Date	Claim / Problem	Evidence Collected	Alternatives Considered	Decision Made	Supervisor Discussion
2026/06/26	Political events influence CSE market behavior, but no unified prediction framework exists	Literature review of Karunarathna & Wijayanayake (2015), Attapattu & Gunarathne (2013), Jayasinghe (2010), Hulangamuwa & Dissanayake (2022); 2024 election market reports from Daily FT	(1) Focus only on event study analysis. (2) Focus only on ML prediction without event study. (3) Integrate event study with ML prediction.	Selected integrated approach combining Event Study methodology (AR/CAR) with MLR and LSTM for weekly prediction	Discussed with Dr. Janaka Alawatugoda, confirmed integrated approach addresses identified research gap
2026/07/03	Need to define scope of election events to include	2024 Presidential election data (Rs. 273B pre-election decline, Rs. 612B post-election surge); 2024 Parliamentary election data; Provincial Council election history 2000-2021	(1) Include only Presidential elections, (2) Include Presidential + Parliamentary only, (3) Include all election types + Cabinet transitions	Include Presidential, Parliamentary, Provincial Council, Local Government elections, and Ministry/Cabinet transitions	Discussed with Ms. Sonali Jayarathne, confirmed comprehensive scope strengthens research contribution
2026/07/10	Selection of appropriate event windows for analysis	Prior research using 80-day windows (Rajarata University study); 2/7/15/30-day windows common in event study literature (MacKinlay 1997; Brown & Warner 1985)	(1) Use single 30-day window, (2) Use multiple windows (2/7/15/30 days), (3) Use custom windows based on event type.	Selected multiple windows (2/7/15/30 trading days) to capture short-term and medium-term market reactions	Discussed with Dr. Emil Kumara, recommended multiple windows for robustness
2026/07/17	Selection of prediction models	Hulangamuwa & Dissanayake (2022) achieved MLR 97.01% and LSTM 87.32%	(1) Use only MLR, (2) Use only	Selected dual-model approach: MLR for interpretability, LSTM for temporal	Discussed with Dr. Janaka Alawatugoda, approved dual-

		accuracy with election influence variable; LeCun et al. (2015) on deep learning capabilities	LSTM, (3) Use both MLR (baseline) and LSTM (temporal)	pattern capture	model approach
2026/07/24	Data availability and quality concerns	CSE market data available from 2009 onwards; Election Commission data accessible; Daily FT archives for 2024 elections; CSE Kandy branch discussion	(1) Start from 1985, (2) Start from 2000, (3) Start from 2009	Selected January 2009 to December 2025 to avoid immediate post-war structural shock while ensuring sufficient event coverage	Discussed with CSE Kandy branch representatives, confirmed data availability for selected period
2026/07/31	Validation strategy for time-series prediction	Walk-forward validation literature; concerns data leakage in financial ML models	(1) Simple train-test split, (2) K-fold cross-validation, (3) Walk-forward validation	Selected walk-forward validation to reduce lookahead bias and simulate real-world prediction conditions	Discussed with Dr. Janaka Alawatugoda, confirmed walk-forward validation is appropriate
2026/08/07	User requirements validation	Survey conducted with 16 respondents (investors, analysts); Domain expert consultations	(1) Literature-based requirements only, (2) Survey-based requirements, (3) Combined approach	Combined literature review, stakeholder discussions, domain-expert consultation, and design-thinking activities	Survey shared with domain expert's channel; feedback incorporated
2026/08/14	Feature engineering for prediction model	Event Study AR/CAR outputs; Election type, stage, political direction variables; Historical CSE data (ASPI, S&P SL20, volume, returns)	(1) Use only market data, (2) Use only event features, (3) Combine event features with market data	Selected combined feature set: AR, CAR, event type, event stage, political direction + historical CSE data	Discussed with Dr. Emil Kumara, confirmed event features add predictive value
2026/08/21	Development methodology	Agile methodology literature; iterative prototyping	(1) Waterfall approach,	Selected Agile with iterative prototyping to	Discussed with supervisor, approved Agile

	selection	benefits for research projects	(2) Agile with iterative prototyping, (3) Hybrid approach	allow refinement of event windows, features, and models based on experimental findings	approach
2026/09/04	Budget and work breakdown structure	Hardware availability; open-source software; cloud service estimates; WBS and Gantt chart requirements	(1) High-cost development setup; (2) Minimal-cost open-source approach; (3) Outsourced development	Selected minimal-cost approach using existing hardware, open-source tools, and limited cloud services	Discussed with supervisor approved budget and WBS alignment
2026/09/11	Finalize evaluation and validation framework	Metrics table, validation strategy, prior studies, SUS guidelines, walk-forward literature, supervisor feedback	(1) AR/CAR only; (2) MAE/RMSE only; (3) multi-metric evaluation	Use multi-metric framework: AR/CAR significance, Directional Accuracy, MAE, RMSE, MAPE, Precision/Recall/F1, walk-forward validation, SUS ≥ 70	Reviewed with Dr. Janaka Alawatugoda, approved alignment with research objectives

Table 15 Mandatory Proposal Evidence and Decision Log

Appendix 3 — AI Use Disclosure

AI Tool	Purpose of Use	What Was Generated/Assisted	How Output Was Verified	Final Student Contribution
DeepSeek	Collected data briefing	Brief explanations of event study methodology, AR/CAR concepts, and political-event market impact from collected sources	Cross-checked with original academic sources and collected data; discussed with supervisor	Student selected all sources, performed critical synthesis, and wrote the final literature review
DeepSeek	Concept clarification	Explanations of MLR, LSTM, walk-forward validation, and evaluation metrics	Verified against academic references and prior CSE studies; discussed with supervisor	Student finalized the methodology, model selection, and validation strategy
DeepSeek	Guidance for brainstorming research gaps and objectives	Provided guidance on possible research gaps and objective direction	Evaluated against existing CSE literature; refined based on supervisor feedback and domain-expert consultation	Student identified the final research gap, main objective, and specific objectives
DeepSeek	Grammar and paraphrasing	Rephrasing of selected sentences and improvement of academic tone	Reviewed manually; accepted only changes that preserved original meaning and technical accuracy	Student maintained ownership of all arguments, interpretations, and final wording
Google Gemini	Document formatting and table structuring	Suggested table layouts, heading structures, and section formatting	Checked against university proposal template and supervisor requirements	Student populated all content, verified accuracy, and finalized the document
DeepSeek	Methodology justification support	Draft wording for methodology justification and evaluation metric descriptions	Cross-checked with academic sources and prior studies; revised for accuracy	Student chose final explanations and ensured alignment with the research design

Google Gemini	Consistency checking	Flagged possible inconsistencies in terminology, abbreviations, and section ordering	Reviewed manually against the full proposal; corrected where necessary	Student made all final editorial decisions and approved the submitted version
DeepSeek, Google Gemini	Reference formatting guidance	Suggested citation formats and reference list organization	Verified against IEEE style guide and original sources	Student compiled, verified, and finalized all references

Table 16 AI Use Disclosure